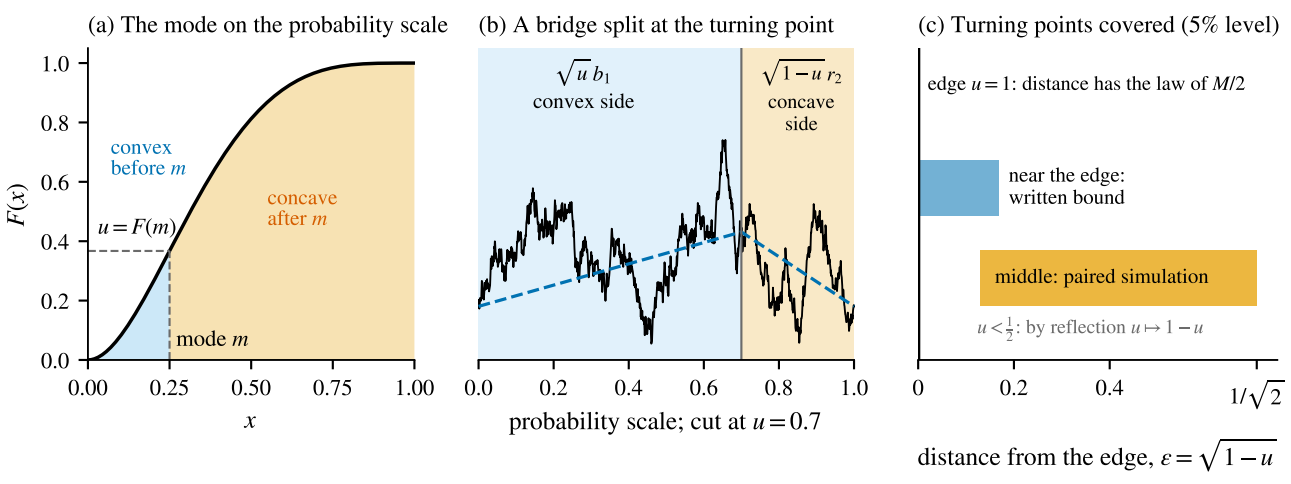
(a) The mode on the probability scale
1.0
0.8
0.6
0.4
0.2
0.0
F(x)
convex
before m
concave
after m
u = F(m)
mode m
0.00
0.25
0.50
0.75
1.00
x
(b) A bridge split at the turning point
√u b1
convex side
√1−u r2
concave
side
0.0
0.2
0.4
0.6
0.8
1.0
probability scale; cut at u = 0.7
(c) Turning points covered (5% level)
edge u = 1: distance has the law of M/2
near the edge:
written bound
middle: paired simulation
u < 1/2: by reflection u ↦ 1 − u
0
0.2
0.4
1/√2
distance from the edge, ε = √1−u